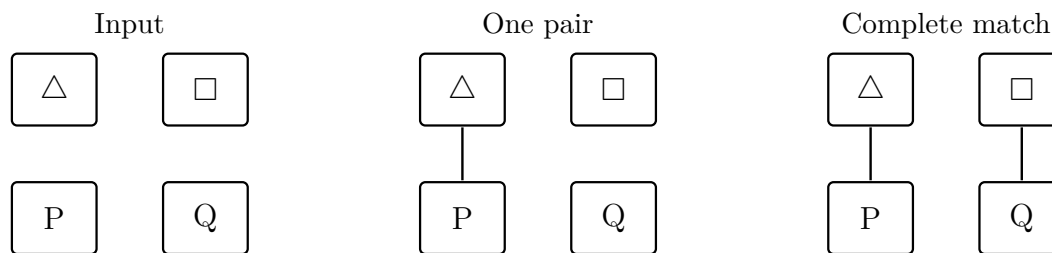


Queues run from left to right. A match gives each customer exactly one partner in the other queue. Nobody is left out. It keeps order when any two customers and their partners appear in the same order.



Problem 1: Find every match between these two queues. Which matches keep order? Use cards and yarn to test your matches.

R1

R2

R3

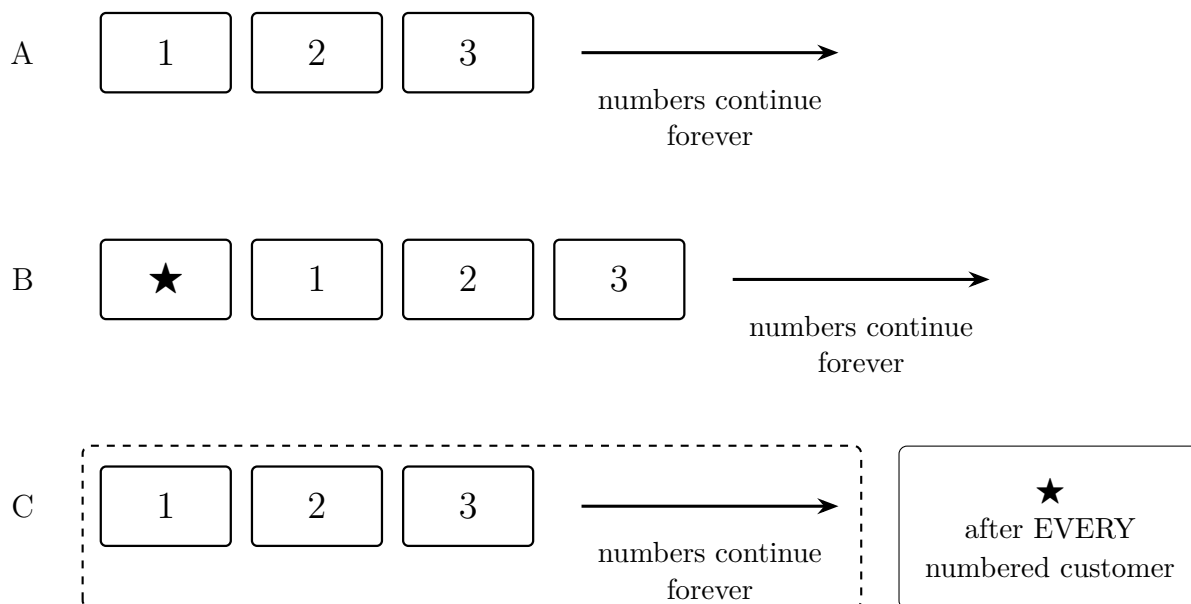
B1

B2

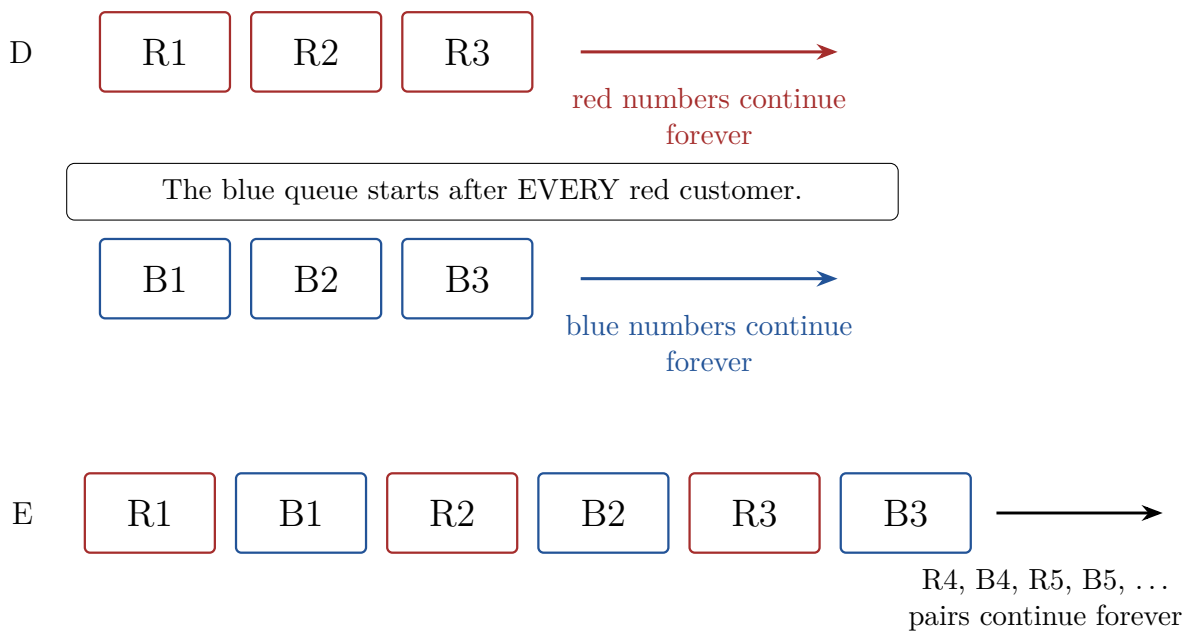
B3

An arrow marked “numbers continue forever” represents all later numbered customers.

Problem 2: Compare A with B, then A with C. Can you match every customer? Can a match keep order? Give a rule for every customer, or explain why it is impossible.

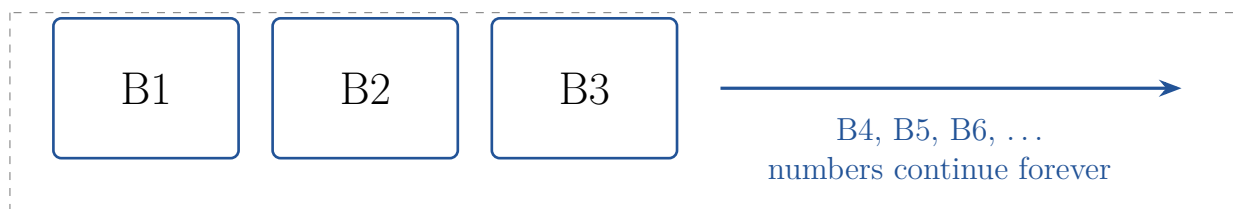
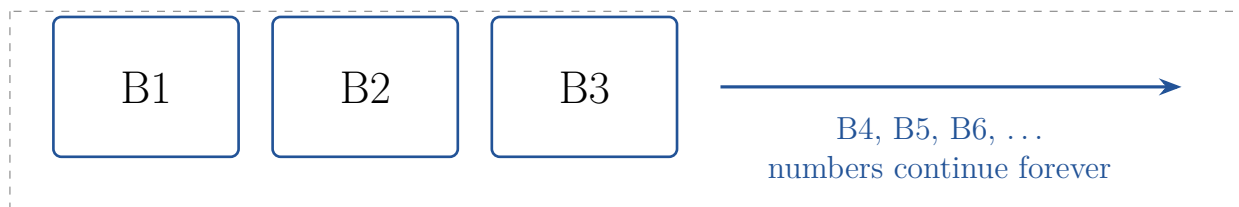
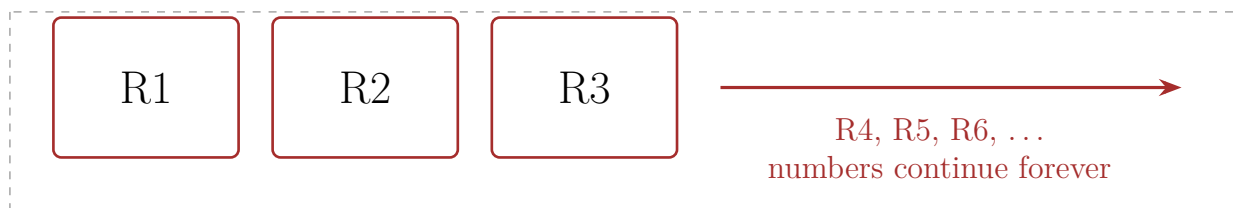
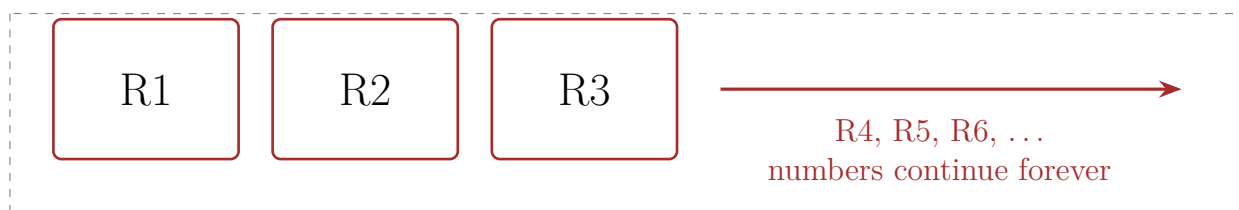


Problem 3: Compare D with E. Can you match every customer? Can a match keep order? Give a rule for every customer, or explain why it is impossible.

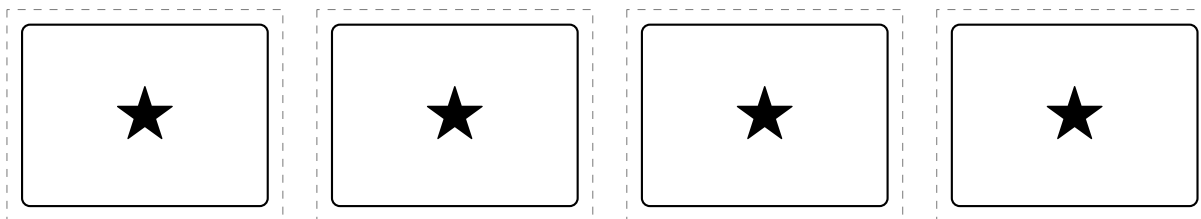


Problem 4: Arrange one or two endless numbered rows and up to two ★ customers on the table. A star or new row following a whole endless row comes after EVERY customer in that row.

Find two different arrangements that can be matched while keeping order. Find two that can be matched only if order may change. Sketch your arrangements in the boxes. Give matching rules or explain the obstruction.



R1	R2	R3	R4
R5	R6	R7	R8
R9	R10	R11	R12
B1	B2	B3	B4
B5	B6	B7	B8
B9	B10	B11	B12



This customer comes after EVERY customer
on the earlier strip.

This customer comes after EVERY customer
on the earlier strip.

This whole row starts after EVERY customer
on the earlier strip.

This whole row starts after EVERY customer
on the earlier strip.